Advay Kadam

732-986-7792 | advayk2@illinois.edu | linkedin.com/in/advay-kadam | advay-k1.github.io/

Education

University of Illinois Urbana-Champaign — Champaign, IL
B.S. Computer Science and Statistics | GPA: 3.9 — *May 2027*
- **Selected Coursework:** Graduate Distributed Systems, Machine Learning Systems, Operating Systems, Deep Generative Models, Applied Parallel Programming, Computer Architecture, Data Structures in C++
- **Awards: Top 3%** at IMC Prosperity (International Trading Challenge), **2x** Dean's List
- **Course Assistant:** Data Structures and Algorithms in C++, Statistical Modeling (Applied Regression)

Experience

Amazon — May 2026 – Present
Software Development Engineer Intern — *New York City, NY*
- Architected and implemented an end-to-end Java service that automates provisioning of internet egress infrastructure for new AWS Region launches, reducing manual deployment effort by **2 months** per region
- Designed and deployed a REST API on AWS ECS Fargate to orchestrate provisioning of Network Load Balancers, PrivateLink endpoints, NAT Gateways, and supporting networking infrastructure
- Engineered infrastructure capacity for upcoming AWS Region launches, provisioning internet egress resources to support **6+ exabytes** of projected network traffic

Supercomputing Systems and AI Lab @ UIUC — Aug 2025 – Present
Machine Learning Systems Researcher (PI: Dr. Minjia Zhang) — *Urbana, IL*
- Developed a fused RMSNorm + GEMM kernel using the Neuron Kernel Interface (NKI) for AWS Trainium, optimizing Transformer inference through architecture-aware scheduling across Tensor, Vector, and DMA engines
- Implemented row-blocked execution to overcome Trainium on-chip memory constraints, enabling efficient inference on sequences up to **32K** tokens while sustaining **82%** Matrix Functional Unit utilization
- Integrated sequence sharding across two NeuronCores, achieving up to **1.5×** end-to-end inference speedup over the PyTorch/XLA baseline

Bank of New York Mellon — Jun 2025 – Aug 2025
Software Engineer Intern — *Jersey City, NJ*
- Built a FastAPI service for a LangGraph/LangChain workflow with **4** agents + orchestrator, generating explainable column-level security labels from dataset metadata
- Integrated AWS SQS to process **500+** daily dataset requests across LOBs, cutting end-to-end latency by **30%** with chunking and bounded concurrency
- Deployed AWS Lambda workers to run asynchronous agent pipelines and persist **30,000+** security labels to S3

Projects

Tapestry: Concurrent Memory System for LLM Agents | *Python, PyTorch, LangChain, Distributed Systems*
- Built a snapshot-isolated memory layer for LLM agents with staged writes, deterministic commits, and checkpointed recovery
- Reduced GAIA benchmark token usage by **63%** and latency by **45%**; implemented SQLite WAL recovery for **8.8×** faster crash recovery

Measuring Effective Context Utilization in LLMs | *Python, PyTorch, Hugging Face Transformers, NumPy*
- Evaluated 6 attribution methods across 3 context lengths and 5 needle depths, profiling peak VRAM, wall-clock latency, and prompt-region influence on an NVIDIA H200 GPU
- Implemented batched attention-mask and substitution ablations, reducing profiling runtime by up to **3.48×** versus physical chunk deletion while preserving RoPE token positions

Hybrid Distributed File System (HyDFS) | *C++, Sockets (UDP/TCP), Multithreading, Linux*
- Built a Cassandra-inspired distributed file system across **10** Linux machines in C++, implementing consistent hashing, UDP gossip membership, TCP file transfer, and 3-way replication

Technical Skills

Languages: Python, C, C++, Java, SQL
ML Frameworks + Libraries: PyTorch, CUDA, LangChain/LangGraph, AWS Neuron Kernel Interface, NumPy
Systems + Tools: Linux, Bash, Git, Docker, Networking (TCP/UDP), AWS (SQS, Lambda, EC2), FastAPI, PySpark